Due Nov 13, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Find the horizontal asymptote(s) of 1 / 1 point

$f(x) = \dfrac{3x^2 + 4x}{2x^2 - 1}$.

Select all of the options below that are horizontal asymptotes.

- [ ] No horizontal asymptote exists
- [ ] $y = 0$
- [x] $y = \dfrac{3}{2}$

Nice work

To find the horizontal asymptote, take the limit of the function as x approaches $\pm\infty$.

$\lim\limits_{x\to\infty} \dfrac{3x^2 + 4x}{2x^2 - 1} = \dfrac{3}{2}$

$\lim\limits_{x\to-\infty} \dfrac{3x^2 + 4x}{2x^2 - 1} = \dfrac{3}{2}$

So since the limit is equal to a number, then a horizontal asymptote occurs at that y-value. Both of the limits are the same, so there is only one asymptote at $y = \frac{3}{2}$.

- [ ] $x = -\dfrac{1}{\sqrt{2}}$
- [ ] $x = \dfrac{1}{\sqrt{2}}$

2. Find the horizontal asymptote(s) of 1 / 1 point

$f(x) = \dfrac{|x|}{2x^2 + 2}$.

Select all of the options below that are horizontal asymptotes.

- [ ] $f(x)$ has no horizontal asymptotes.
- [x] $y = 0$

Nice work

To find horizontal asymptotes, find the limit of the function at positive and negative infinity.

$\lim\limits_{x\to-\infty} f(x) = \dfrac{|-\infty|}{2(-\infty)^2 + 2} = 0$

$\lim\limits_{x\to\infty} f(x) = \dfrac{|\infty|}{2\infty^2 + 2} = 0$

So the horizontal asymptote occurs at $y = 0$.

- [ ] $x = \dfrac{1}{2}$
- [ ] $y = \dfrac{1}{2}$
- [ ] $x = -\dfrac{1}{2}$

3. Find the horizontal asymptote(s) of 1 / 1 point

$f(x) = 3x + 2$.

Select all of the options below that are horizontal asymptotes.

- [ ] $x = -\dfrac{2}{3}$
- [ ] $y = 3$
- [ ] $x = \dfrac{2}{3}$
- [x] $f(x)$ has no horizontal asymptote.

Nice work

To find the horizontal asymptote, take the limit of the function as x approaches $\pm\infty$.

$\lim\limits_{x\to\infty} 3x + 2 = \infty$.

$\lim\limits_{x\to-\infty} 3x + 2 = -\infty$.

So since the limit is equal to infinity, we say that the horizontal asymptote does not exist.

- [ ] $y = 2$

4. Find the limit. 1 / 1 point

$\lim\limits_{t\to\infty} \dfrac{t - t\sqrt{t}}{2t^{3/2} + 2t - 5}$

- ○ 0
- ◉ $-\dfrac{1}{2}$
- ○ ∞
- ○ $\dfrac{1}{2}$

Nice work

Good work.

Use the fact that $\lim\limits_{x\to\infty} \dfrac{1}{x^r} = 0$ when r is positive. Divide the numerator and denominator by the highest power of t that appears, which is $t^{3/2}$. Then simplify.

$\lim\limits_{t\to\infty} \dfrac{t - t\sqrt{t}}{2t^{3/2} + 2t - 5} = \lim\limits_{t\to\infty} \dfrac{\frac{t - t\sqrt{t}}{t^{3/2}}}{\frac{2t^{3/2} + 2t - 5}{t^{3/2}}}$

$= \lim\limits_{t\to\infty} \dfrac{\frac{1}{t^{1/2}} - 1}{2 + \frac{2}{t^{1/2}} - \frac{5}{t^{3/2}}} = \dfrac{\lim_{t\to\infty} \frac{1}{t^{1/2}} - 1}{2 + \lim_{t\to\infty} \frac{2}{t^{1/2}} - \lim_{t\to\infty} \frac{5}{t^{3/2}}}$

$\dfrac{0 - 1}{2 + 0 - 0} = -\dfrac{1}{2}$

5. Find the limit. 1 / 1 point

$\lim\limits_{x\to\infty} \sqrt{x^2 + 1}$

- ○ 0
- ○ does not exist
- ◉ ∞
- ○ 1

Nice work

Good work.

When x becomes infinitely large, so does $x^2 + 1$, and thus so does $\sqrt{x^2 + 1}$. Thus this limit is equal to $+\infty$.

Note that both $x^2 + 1$ and $\sqrt{x^2 + 1}$ are always positive, so the same applies when x goes to $-\infty$

6. Find the limit. 1 / 1 point

$\lim\limits_{x\to\infty} \left[e^{-x} + \cos\left(\dfrac{x}{2}\right)\right]$

- ◉ does not exist
- ○ 1
- ○ ∞
- ○ 0

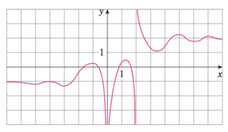
Nice work

Correct.

$\lim_{x\to\infty} e^{-x} = 0$. However, the function $\cos\left(\frac{x}{2}\right)$ cycles between -1 and 1, so $\lim_{x\to\infty} \cos\left(\frac{x}{2}\right)$ does not exist.

Therefore, the limit of the sum of these two functions at ∞ does not exist, either.

7. The function $g(x)$ is graphed below. 1 / 1 point



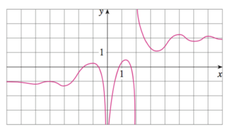
Find $\lim\limits_{x\to\infty} g(x)$

- ○ 0
- ○ does not exist
- ○ ∞
- ◉ 2

Nice work

Good work. Look at the right side of the graph. As the x-values increase, the graph gets closer and closer to $y = 2$.

8. The function $g(x)$ is graphed below. 1 / 1 point



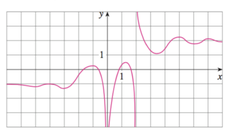
Find $\lim\limits_{x\to-\infty} g(x)$.

- ◉ -1
- ○ $-\infty$
- ○ 0
- ○ ∞

Nice work

Good work. Look at the left side of the graph. As the x-values decrease, the graph gets closer and closer to $y = -1$.

9. The function $g(x)$ is graphed below. 1 / 1 point



Find all **vertical** asymptotes of g.

- [ ] $x = -1$
- [ ] $y = 2$
- [ ] The function has no vertical asymptotes.
- [x] $x = 2$

Nice work

As x approaches 2 from the left, the function values go to $-\infty$. As x approaches 2 from the right, the function values go to ∞.

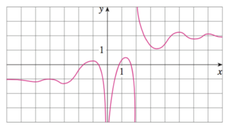
- [ ] $y = -1$
- [x] $x = 0$

Nice work

As x approaches 0 from either the right or the left, the function values go to $-\infty$.

- [ ] $y = 0$

10. The function $g(x)$ is graphed below. 1 / 1 point



Find all **horizontal** asymptotes of g.

- [x] $y = 2$

Nice work

As the x values increase, the function gets close to the line $y = 2$. Therefore $y = 2$ is a horizontal asymptote.

- [ ] $x = -1$
- [ ] $x = 2$
- [ ] $y = 0$
- [ ] The function has no horizontal asymptotes.
- [ ] $x = 0$
- [x] $y = -1$

Nice work

As the x values decrease, the function gets close to the line $y = -1$. Therefore $y = -1$ is a horizontal asymptote.